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APPLE AND THE MUSIC INDUSTRY[1]

On November 1, 2018, a headline in the *Financial Times* announced, "Apple to stop revealing unit sales sparking peak iPhone fears," and investors sold stock in Apple Inc. (Apple), pushing its value down by almost 7 percent.[2] Observers wondered if slowing hardware sales would spell trouble for Apple's position in the streaming music industry, where despite its dominance in hardware, its brand image, and its large cash cushion, the company continued to lag behind market leader Spotify AB (Spotify) in terms of its number of paid subscribers.[3] Could Apple reinvent itself again in the music industry, as it had done in the past?

Apple had once been a maverick in the global music industry, striking a deal to distribute downloadable music from its iTunes Store and growing market share with the launch of its iPod in 2001. The iPod had at one point had a 92 per cent share of the music player market,[4] before competitors such as Microsoft Corporation (Microsoft) and SanDisk entered the industry and drove the iPod's share down to 71 per cent.[5] Then the company launched the iPhone in January 2007, reigniting growth in the sale of digital music downloads.

But then the disrupter was disrupted. The emergence of streaming music in 2005 seemed to take the incumbents by surprise. By 2019, streaming was the largest segment in the music industry by revenues, accounting for US$8.9 billion[6] in sales out of a $19.1 billion global market. There were new players, including Spotify, Amazon.com Inc. (Amazon), Google LLC, and Pandora.[7] It seemed that change was the only constant in the global music industry. Apple had led the initial break from the past and had profited immensely from it. What would Apple need to do to continue to play a leading role in the music industry's future?

THE MUSIC RECORDING INDUSTRY

Recorded music, as a form of entertainment, could be encountered anywhere in North America and, indeed, in much of the world. Thousands of radio stations broadcast song lists interspersed with commercials; televised music videos were available on music channels such as MTV; and public venues such as shopping malls, bars, and restaurants played background music. Music tastes among consumers spanned a variety of options—from classic rock, heavy metal, rap, and country to other existing and emerging genres. Although only a fraction of the countless recording artists achieved success and fame, the perceived glamour of the music industry continued to draw large numbers of aspiring recording artists, all eager to become the next star.

Half of all Americans purchased recorded music, in the form of compact discs (CDs) and other media, at least once a year. Sales in the US music industry were $19.1 billion in 2018 (see Exhibit 1).[8]

Before 2000, the vast majority of music industry revenues had come from the sale of physical units in retail stores, but music sales were declining in the early 2000s, going from $14.3billion in 2000 to $12.3 billion in 2004.[9] Record label executives believed that a large portion of this decline in physical unit sales was due to online piracy. Recording Industry Commercial Piracy Report estimated that global piracy cost the recording industry more than $4 billion a year.[10] Observers countered by saying that, as record labels had become more cautious in signing artists, the number of new albums released each year had dropped since its peak in 1999.[11] In addition, the record labels were slow to manage the marketing and distribution costs associated with launching a new album, and investors considered the recording industry to be a high-cost, high-risk, low-return business.

In the early 2000s, the recording companies—the key group that coordinated the development and distribution of recorded music—included four major record labels: Sony BMG (Sony), Warner Music Group Inc. (Warner), Universal Music Group, and EMI Group Limited (EMI). Combined, these recording companies produced and distributed almost 88 per cent of all music recorded,[12] Record labels (or recording companies) signed up recording artists, developed, produced, marketed, and often distributed the recordings. By the time a consumer purchased a music CD at a retail outlet, that product had passed through several steps (see Exhibit 2), with various royalties and revenues accruing at different points in this journey (see Exhibit 3). Since only 10 per cent of all artists were profitable—in the sense that their music shipments, net of returns, generated revenues higher than the costs to produce the music—the four major labels were constantly on the lookout for the next chart-topping recording artist and for fast-growing, independent recording labels, whose acquisition would add to their growth.

Thousands of independent labels accounted for the balance of the music recording industry. Maturing computer technology—advanced music recording software and hardware; high-powered yet low-cost computers; and inexpensive recording media, such as CDs—allowed budding recording artists to develop and release independent CDs, often self-marketing them through Internet websites.

Distributors typically sold product from record labels to retail chains, earning an estimated 10 per cent of the retail price. In the United States, mass merchandisers such as Wal-Mart and Target accounted for 65 per cent of retail music CD sales, with the rest being sold in specialty and online stores.[13] In the United States, as recorded music was a small proportion of mass merchandisers' total sales, retailers generally demanded price reductions of 1–2 per cent per year. These mass merchandisers had limited selections of music, choosing to stock popular releases that appealed to wide audiences. Highly concerned with inventory turnover, mass merchandisers generally did not carry a deep selection of music in any genre. Over the past decade, online retailers had begun to fill this gap, though they accounted for no more than 1 per cent of the industry's CD sales.

Consumers purchased CDs at traditional retailers, such as music specialty stores and mass merchandisers, and through online retailers. Although production costs had dropped since CDs were first introduced in the 1980s—blank CDs could now be purchased for less than a dollar each—a typical music CD continued to retail at about $15. Industry observers believed that the drop in production costs had been offset by growing marketing and promotional costs related to CD launches. A wider variety of music CDs was available at specialty retailers than through mass merchandisers. At the retail level, consumers had no choice but to purchase an entire CD, even if they wanted only a single song from the 10-song playlist.

Hardware advances had enabled the development of segments outside of home entertainment. The launch of Sony's cassette tape–playing Walkman in the late 1970s offered consumers a portable device. The subsequent proliferation of portable music devices increased the selection available and reduced prices, which drove demand for both hardware and music recordings. Until 1998, digital music files could be played only on computers; however, the launch of Diamond Multimedia's Rio 300 MP3 player allowed consumers to carry with them and play 16 digital songs. Within a few months of Rio's introduction, the number of companies offering MP3 players grew into the hundreds. Within a year, the digital storage capacity for MP3 players increased from about one hour to five hours (or about 60 songs). However, there was no set standard for design with regard to placement and sequence of device buttons, nor were there limits to the number of functions a manufacturer could include. The result was that functionality varied significantly from device to device.

Music format technology continued to advance at a rapid pace. Newer formats such as CDs were cross-compatible with other systems (for example, music CDs were playable on DVD machines) and could store greater amounts of data than old formats such as cassette tapes or vinyl long-play records (LPs). The newest physical format, the DVD, could hold more data than CDs. Digital formats, such as MPEG, and its later versions, such as MP3 files, increased the ease by which music could be shared. Before the recent arrival of encryption technology, songs could be easily copied or "ripped" from CDs, converted into digital format, then e-mailed as attachments to many recipients. With older formats such as the cassette tape, it was tedious for consumers to make more than one physical copy of a tape at a single sitting.

The equipment necessary to duplicate CDs ranged from simple home computers with CD burners to high-volume, commercial-grade machines, and both were easily obtainable. CD covers could be easily scanned and printed on high-quality colour printers, giving illegal copies a professional look. Although pirated copies of CDs were usually sold by street hawkers or through an individual's network, in some countries, there were large-scale operations dedicated to the illegal copying and distribution of CDs. Pirated CDs could sell for less than 10 per cent of the retail cost of the real thing.

Although piracy was thought to account for billions of dollars' worth of lost industry revenues annually, piracy itself created more noise than sales losses. However, music executives noticed that, when digital formats became available to consumers, consumers were more interested in downloading music for free than paying for CDs, whether authentic or pirated. The Recording Industry Association of America (RIAA), of which the four major recording labels were members, became increasingly concerned about the ease with which songs could be shared online.[14] Although some music was digitally recorded for free distribution, a large amount of digital music in circulation contained pirated music. The RIAA began taking action, suing websites such as MP3.com for failing to pay royalties on the music it made available to consumers.[15] Digital player manufacturers such as Diamond Multimedia were also sued.[16]

In addition to their legal manoeuvres, music industry members also attempted to develop standards to combat illegal downloads. In 1998, the Secure Digital Music Initiative (SDMI) was founded by members of the music recording industry to develop secure methods for distributing digital music. The SDMI's resulting plan encouraged the launch of SDMI-compliant music players that would play SDMI-compliant content only, in Phase 1, and computer applications that would prevent consumers from obtaining or distributing music on illegal sites, in Phase 2.[17] The SDMI even ran a "Crack SDMI" contest, offering $10,000 as a prize for hackers who could successfully defeat SDMI's screening technology. Several hacker groups called for a boycott of the contest, saying record labels were trying to limit consumers' "fair use" rights to make copies of music they purchased for their own personal use in a car stereo or laptop computer. To SDMI's dismay, barely a month after the contest was launched, hackers successfully defeated the SDMI systems.[18]

The RIAA's biggest challenge came in 1999, when a then-unknown university student decided to code and release a peer-to-peer (P2P) file-sharing program called Napster.

NAPSTER AND THE RIAA

When a roommate complained about the difficulty of finding downloadable music on the Internet, Shawn Fanning, a freshman at Northeastern University, in Boston, developed a program to change the way people shared music online. His program, Napster, allowed computers to link up with each other and trade files over the Internet. Fanning made Napster available as a free download, and the program acted as a central hub—a P2P network that allowed users to exchange files with each other, particularly music files. Traffic passed through Napster's centralized servers, but no copies of the music files were saved on Napster's computers. When he released Napster, Fanning had no intention of profiting from it but saw it only as "a cool way to build community."[19]

By 1999, the Internet was rapidly gaining popularity, especially among the teenage population, who swiftly adopted tools such as email. News of Napster spread, especially in college dormitories, since students typically had the high-speed Internet access necessary to download large files quickly.

Even with Napster, downloading music on the Internet was difficult for those who were not technologically inclined. To download a song, a user had to run a P2P program to search for a song, which would take 10 seconds to two minutes, depending on the popularity of the track. After the program had found other connections on the P2P network that the user could connect to and download music from, the user could download an individual file from one of the available connections. However, the user faced several risks: the track might be corrupt—that is, it could not be played once it had been downloaded; the user might receive an incomplete file or the wrong version; or they might receive a virus-contaminated file.

Users also required fast Internet connections to download the songs at a reasonable speed: a high-speed connection would be able to download a multi-megabyte song within minutes, but a slower dial-up Internet connection could take anywhere from five minutes to half an hour to download the same file. In addition, the other user (from whose computer the user was downloading the song) could unexpectedly shut off the connection by logging off, requiring the user to go back into the system and search for another reliable connection from which to download.

Despite the difficulties involved, by late 1999, millions of users had downloaded from Napster, and according to the company, the user base was growing at between 5 per cent and 25 per cent daily.[20] Concerned at the rising amount of Internet bandwidth taken up by students using Napster to trade files, some US colleges moved to block the service altogether.

Fanning's uncle, realizing the potential of Fanning's venture, approached venture capitalists and secured seed funding for Napster. Fanning then dropped out of school and moved west to continue working on his program. Napster's plan was to earn revenues through user membership fees, with a portion of the earnings remitted to the recording industry.[21]

In December 1999, the RIAA, which represented the US recording industry, including all the major record labels, sued Napster for copyright infringement. The RIAA—of which Napster was a member—claimed that Napster facilitated piracy. Knowledge of Napster, already spreading quickly, was given a boost because of the publicity from the lawsuit. Almost immediately, downloads from Napster jumped into the millions per week. During the trial, it was revealed to the court that Napster's goals, as contained in early business plans, were "to brin[g] about the death of the CD" and to "bypass the record industry entirely."[22]

By February 2001, Napster was ordered by the US Court of Appeals to stop the distribution of copyrighted materials. The court ruled that Napster's service was not protected by fair use and that Napster was guilty of two kinds of copyright infringement (contributory and vicarious). An article on the ruling noted that Napster had "failed to police its system in an attempt to stop the spread of copyrighted works," and had inflicted "substantial harm to recording companies. 'Napster, by its conduct, knowingly encourages and assists the infringement of [record company] copyrights' the court wrote in its decision."[23]

Napster responded by blocking the download of more than 250,000 songs by screening for 1.6 million song titles, but users found a way around this problem simply by generating new names for songs.

Unfortunately for Napster, the ongoing legal fight was depleting its cash reserves. In early 2000, Napster executives rethought their company's strategy, considering cash-generating ventures such as selling advertising, adding e-commerce capabilities to sell concert tickets, or even pursuing co-promotional opportunities with record companies.[24] Napster was then ordered by the court to stop the exchange of all music on its servers until it could block all music owned by other music companies. This order effectively shut Napster down. Napster's name and assets were eventually sold to the software company Roxio.[25]

Meanwhile, two major alliances in the music industry attempted to launch their own online music distribution sites: MusicNet (which included Warner, BMG, EMI, and RealNetworks) and PressPlay (a partnership of Universal Music Group and Sony Music Entertainment with Microsoft, which provided digital media technology and MSN Internet service). While the US Department of Justice launched an antitrust investigation to determine whether the alliances were anti-competitive, music consumers largely dismissed both services.[26]

Napster's notoriety drove other programmers to develop their own file-sharing programs: LimeWire, Gnutella, BearShare, Morpheus, Grokster, and Kazaa, to name just a few. The programmers were aided by the availability of the base software code, which had been released by hackers. By 2003, at least 14 different file-sharing software programs were freely available. Politicians contemplated bills to outlaw P2P programs, and the RIAA continued its tactics, suing Napster clones in the courts.

Unlike Napster, however, these new file-sharing programs did not require centralized servers to manage traffic. Instead, they allowed peers to connect directly with other peers, forming a decentralized network of individual PCs with no centre. Thus, Kazaa, for example, could not be shut down in the same way that Napster had been forced to shut down—by switching off a set of centralized servers.

The RIAA started using a different tactic, employing software that sought out pirated music in the file-sharing networks so as to gather evidence against individuals. While it was not illegal for consumers to download music or to make copies for their own personal use, the RIAA argued that making music available for download, en masse, was a violation of the music owners' copyright. In a well-publicized series of more than 250 lawsuits, the RIAA sued and won damages against individuals who were making more than 1,000 songs available to peers on their computers.[27]

In addition, the RIAA hired small independent companies called "spoofers" to dilute file-sharing networks with fake music files. These companies set up accounts on the major file-sharing networks and shared files that were designed to look like actual music files but were either totally blank or included only partial songs.

The president of one spoofing company, MediaDefender, stated that the idea was "to frustrate users who are trying to download copyrighted songs."[28]

In addition to the music industry, the film industry was another target of file-sharing consumers. On June 27, 2005, in what was considered a win for both industries, the US Supreme Court ruled unanimously, in *Metro-Goldwyn-Mayer Studios v. Grokster Ltd.*, that file-sharing companies could be liable for copyright infringement if their products encouraged consumers to illegally swap songs and movies. However, observers noted that, even if file-sharing program companies such as Grokster collapsed, their software was already installed and being used by millions of consumers. Other companies not named in the case continued to offer new types of file-sharing software with more sophisticated features, such as the capability to handle larger files. Finally, independent programmers were likely to continue to develop file-sharing applications and quietly release them online.[29]

LEGAL GUIDELINES CONCERNING COPYRIGHTED MUSIC

By maintaining an information website, the RIAA attempted to educate consumers about copyright laws as they pertained to the evolving and emerging case of digital music. The courts had concluded, for example, that it was legal to download music from websites authorized by the owners of copyrighted music. Also legal was copying music to an analog cassette tape, if the tape was not to be used for commercial purposes. Third, music could be copied to special audio CD-Rs, mini-discs, and digital tapes if royalties had been paid on the media. Again, these copies could not be used for commercial purposes. Beyond that, the RIAA suggested there was "no legal 'right' to copy the copyrighted music on a CD onto a CD-R" and it was illegal to upload or download copyrighted music, give away CD copies, or lend original CDs to others for copying. The organization's guidelines on piracy elaborated further:

> However, burning a copy of a CD onto a CD-R, or transferring a copy to a computer hard drive or your portable music player, won't usually raise concerns so long as:
> - The copy is made from an authorized original CD that you legitimately own
> - The copy is just for your personal use.[30]

APPLE INC.

In 2019, Apple was well known as the maker of the popular iPhone and as the world's first trillion-dollar company (a status it achieved in August 2018),[31] and it was also known for its "walled garden" ecosystem of software, among other things. However, in 2001, Apple was best known as the maker of a niche computer that had largely ceded the desktop operating system war to Microsoft Windows. In 1997, when Dell Computer Corporation CEO Michael Dell was asked what he would do if he were put in charge of Apple, he answered, "I'd shut it down and give the money back to the shareholders."[32] This was a firm that had come back from relative obscurity, launching a series of colourful iMac computers in the late 1990s to revive its brand.

In 2001, Apple had less than 3 per cent of the worldwide personal computer market share, trailing PC (97 per cent).[33] Apple was faced with the challenge of trying to grow its loyal installed user base in a highly saturated and competitive market.

With 25 million users of its personal computer the Macintosh (Mac), worldwide, Apple had an iconic consumer brand, a loyal customer base, and a reputation for developing innovative products.[34] Apple claimed to produce the entire computing package for a user, which, in turn, allowed it to charge higher margins on its products.

The Launch of iPods

In 2001, Apple released the Apple iPod, a hard-drive-based music player with a retail price tag of $399. (Apple's gross margin was 64 per cent of the retail price.) Unlike the typical high-definition, full-colour, full-screen devices of 2019, the first iPod had a small black-and-white display in the top section of a white plastic body. Still, it was capable of storing approximately 1,024 digital songs[35]—a capacity almost 21 times greater than that of the leading MP3 player on the market. Along with the launch of iPod, Apple created an iPod Lounge website, where users could create user profiles; make their own iPod accessories, such as iPod covers; and submit suggestions and ideas regarding iPod changes. The forum was moderated by Apple employees, who sifted through suggestions and forwarded the best ideas to the company.

Apple enlisted a few of the consumers who had been sued by the RIAA to star in television commercials supporting the launch of iPod. The commercials featured dancing silhouettes wearing iPod's distinctive white headphones. Industry observers commented that the iPod was on its way to achieving iconic cult status, and many loyal Apple users considered the iPod to be as Windows users' first introduction to the "Apple way of life." The company also released Apple iPods by Hewlett-Packard and Dell—with the only difference being a small company logo on the back of the Apple iPod below the Apple logo.

Apple widened consumer choice by continually launching variants on the iPod, such as the iPod mini (with a smaller-sized hard drive), the iPod nano (with a flash memory), and the iPod shuffle (an iPod without a screen or click wheel input). Each new generation of iPod typically featured additional refinements, a lower weight, and a smaller size. Notable changes included the touch-sensitive click wheel, which replaced a mechanical scroll wheel; the use of colour displays; and the addition of flash memory models. In addition, special limited-edition models were launched in collaboration with recording artists, athletes, and awareness campaigns to heighten consumer interest.[36] Unit sales of the iPod grew steadily until 2010, and by 2011, they were exceeded by far by unit sales of the iPhone (see Exhibit 4).

The Launch of iTunes Music Store

Before the launch of the iTunes Music Store, Apple CEO Steve Jobs had to convince reluctant recording producers and artists to distribute their music online. Jobs personally demonstrated the music service to key recording artists, emphasizing the elegance and simplicity of the iTunes Music Store and indicating that a credit card would be all that was required to purchase a song or album. He also emphasized his belief that consumers wanted to "own" the music they downloaded.[37]

The iTunes Music Store was launched in 2003, approximately two years after the iPod was first released. The store began with 200,000 songs available to download, each at a cost of $0.99. Analysts were divided as to the profitability of iTune's digital downloads. While some believed Apple earned one cent per song in gross profit, others believed that the costs of running the iTunes Music Store could well exceed the $0.99 per song that Apple received.[38]

Over the years, the iTunes Music Store grew in popularity. By February 2006, more than one billion songs had been downloaded from the store,[39] and in September 2006, iTunes became the first digital music store among the top five sellers of music in the United States (it was fifth overall). Its catalogue of downloads was broadened from music to include television shows, podcasts, video podcasts, and games.

Throughout his negotiations with the music executives, Jobs touted the inclusion of Apple's digital rights management software. Still, it took 18 months for Jobs and a small team of Apple employees to negotiate deals with the four major labels. As Hilary Rosen put it, "If it weren't for Steve Jobs' persistence, I don't think this would have happened."[40]

The Launch of iTunes Software and FairPlay, a Digital Rights Management System

The new iTunes software was created concurrently with the launch of iPods in 2001 and bundled with each Apple iPod purchased by users. It was the only program that enabled users to play songs on their Apple iPods, and it used the company's proprietary FairPlay digital rights management (DRM) system. DRM systems allowed digital content vendors to control how material could be used on any electronic device on which they were installed. Apple's DRM system worked by encrypting (or "wrapping") each audio file to make it compatible with Apple iPods, and the system prevented songs from being traded on illegal P2P networks. Standard digital formats such as MP3 could not be played on the iPod; a user had to rely on the iTunes software to convert MP3 files into Apple's chosen format, advanced audio coding (AAC), to play songs on the iPod.

In addition to encryption, iTunes also organized the songs on users' iPods into categories through features such as playlists and randomly auto-filled iPods from users' personal digital libraries.

Apple's DRM was not infallible. After the launch of the iTunes Music Store, several independent programmers were able to circumvent or remove the encryption from FairPlay-protected files. In 2005, Jon Johansen, a computer programmer, announced that he had developed a new program that stripped the DRM restriction from iTunes, allowing users to download and trade iTunes songs as if they were regular MP3 files.[41] Within a week, Apple announced that it had modified iTunes software to plug the breach in security.[42] In October 2006, Johansen announced that he had reverse-engineered FairPlay to allow Apple competitors to play their proprietary DRM-protected music and movies on iPods without having to work with Apple for permission.[43]

APPLE FACES COMPETITORS

In the digital download market, new competitors were offering innovative pricing structures: Roxio, which had purchased the Napster name and assets, launched what would eventually become known as the music streaming industry. In the fall of 2003, Roxio relaunched Napster as a legal online music downloading service, charging a monthly fee of $15.00 for unlimited access to songs. Instead of asking users to pay per song, Napster allowed consumers to fill portable digital music players with an unlimited amount of music for a monthly fee.[44]

In 2007, Napster allowed registered users to download every track in its two-million-song catalogue to their PCs and to listen to each track up to three times at no charge. (Registration was free.) To transfer an unlimited number of tracks to a compatible portable player (i.e., not the iPod), registered users had to become subscribers, at a cost of $14.95 a month. Subscribers could also purchase individual tracks at a cost of $0.99 each. Roxio spent four years trying to grow the new Napster service. In trying to convince users that paying a flat monthly fee was preferable to $0.99 a song, Roxio even ran an ad, during the 2005 Super Bowl, entitled "Do the Math." The service was unsuccessful, and Napster was sold to Best Buy in the fall of 2008.[45]

Other online music download contenders included Motorola Inc., which announced plans for a mobile cellular phone that offered iTunes-type software; Yahoo! Music Unlimited; and Microsoft's MSN Music, which operated from 2004 to 2006. Due to poor demand, Microsoft announced in November 2006 that it would shut down MSN Music and redirect visitors either to its Zune Marketplace website[46] (Zune was also the brand name for Microsoft's new 30 GB portable music player, selling for $249.99) or to RealNetworks' Rhapsody websites.[47] Microsoft's Zune Marketplace website offered subscribers access to more than two million songs for $14.95 per month. Microsoft allowed a prospective subscriber to listen to songs for three days or three plays, whichever came first, after which the songs would expire unless the user purchased a subscription. A key differentiator between Zune and the other music players was the Wi-Fi connection that allowed users to share songs between Zune units wirelessly.[48]

APPLE LAUNCHES THE IPHONE

Apple's executives initially refused to speculate about the iPod's prospects outside of digital music. The introduction of podcasting, however, apparently caused Apple to change its mind. Podcasting—the term was derived from a combination of *iPod* and *broadcasting*—allowed users to download radio or television station broadcasts and listen to them at their leisure. By 2005, any podcaster (broadcast provider) was able to upload content to the iTunes Music Store.[49]

In January 2007, Apple launched its iconic iPhone, and dropped the word *computer* from its new name, Apple Inc. While other mobile phones at the time had physical keyboards and pre-loaded software, the iPhone featured a touch-screen interface and allowed consumers to customize their devices with apps from Apple's App Store. The popularity of the iPhone allowed Apple's iTunes to become the largest music retailer in the world by 2010. By 2013, iTunes had 435 million users in 119 countries and had sold 25 billion songs.[50]

Sales of iPhones were 1.39 million units in 2007 and rose to a high of 231.22 million units in 2015.[51] In comparison, sales of iPods went from 51.63 million in 2007 to 14.38 million in 2014.[52] There was no official figure for iPod sales in 2015 because, in January of that year, Apple stopped reporting iPod unit sales. Going forward, iPods would be lumped into the "Other Products" category.[53]

THE MUSIC STREAMING INDUSTRY

The growing prominence of digital downloads in the music industry coincided with a rapid decline in the sale of physical music products. From $23.8 billion in 2001, global sales of music declined to $14.2 billion in 2014. The downward trend was reversed by the emergence of a new source of revenues: streaming music. From about $400 million in 2010, streaming revenues rose to $1.9 billion by 2014.[54] Four years later, in 2018, the popularity of streaming boosted global music industry revenues to $19.1 billion, with streaming accounting for $8.9 billion of that total.[55]

Streaming was the latest business model disruption in the industry, and it led to a decline in the number of digital downloads. There were an estimated 60.4 million paying users for music streaming in 2019 alone. In the US, revenue from streaming services more than doubled to $8.8 billion from 2016. In the US alone, 100 million listeners together streamed more than 500 billion songs.[56] Alongside incumbents such as Apple, new competitors were entering the market.

Apple Music

Launched in June 2015, Apple Music charged prices between $4.99 per month and 14.99 per month—with a three-month free introductory period—and gave customers access to a catalogue of 50 million songs.

While it did not offer a free, ad-supported version, Apple was able to convert 80 per cent of the people who tried the service into monthly active users and paid 73 per cent of its revenues, or $12.10 per 1,000 streams, to record labels.[57] The service was available on all Apple devices and non-Apple personal computers and Android phones. Apple Music, which differentiated its service by offering playlists curated by other users, reached 50 million paid subscribers in May 2018.[58]

Spotify

The market leader in the streaming industry was Spotify, launched in 2008. It had a catalogue of more than 30 million songs, which included exclusive live recordings. A popular feature was its computer-generated playlists—Release Radar and Discover Weekly—which were customized to user preferences and introduced users to new music.

Spotify had become the leader by offering a two-tiered service. The free, advertisement-supported version of its service was used by 54 per cent of Spotify listeners. The ad-free Spotify Premium service cost $4.99 to $14.99 a month. By the second quarter of 2019, Spotify had 180 global monthly active users, of which 83 million were paying subscribers.[59]

Amazon Music, SoundCloud, YouTube Music[60]

In 2007, Amazon launched Amazon MP3, an online music store selling digital downloads. While the company continued to make digital downloads available, it also launched Amazon Music Unlimited, a streaming service, in April 2017; this service cost $9.99 a month or $7.99 a month for Prime members.

SoundCloud, launched in 2008, aimed to connect music artists with fans. Its service cost $4.99 or $9.99 per month, with the $9.99 service tier featuring 30 million tracks for listeners. Unlike other streaming services, SoundCloud featured an open platform, where artists could use software tools to market themselves directly to users. SoundCloud charged independent artists $70–$100 for these music creation tools.

YouTube was the most popular online platform for audio entertainment. YouTube Music, launched in May 2018, offered a free, ad-supported version and a $9.99 per month YouTube Music Premium service that was similar to those of the market leaders.

PREPARING FOR THE NEXT PHASE OF GROWTH: APPLE MEDIA?

Apple Music was second only to Spotify in terms of the number of paying subscribers. Apple Music had a market share of 20 per cent in 2018, suggesting that there was room to grow. Looking more broadly, analysts estimated that there was a total addressable market of $24 billion dollars in 2019. Further, Spotify with 35 per cent share was the market leader. Other top players include Amazon (11 per cent), Tencent Holdings Limited (10 per cent) and Youtube (5 per cent).[61]

With Apple's new focus on generating revenue growth from services instead of hardware, it remained to be seen how the company's executives would position it for growth in the global music industry. Apple's array of services included Apple Music, Apple Pay, the news subscription service Apple News+,[62] and the video subscription service Apple TV Plus. It was estimated that Apple Music and Media revenues would rise steadily from 2017 to 2025 (see Exhibit 5). Would the next step be to launch an Apple Media subscription service that bundled all of its media offers for one flat monthly fee?

EXHIBIT 1: GLOBAL RECORDED MUSIC REVENUES, 2001–2018

(In US$ billions)	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010
Physical	23.3	21.7	20.0	19.4	18.1	16.4	14.2	12.0	10.5	9.0
Digital (excluding streaming)	-	-	-	0.4	1.0	2.0	2.7	3.4	3.7	3.9
Streaming	-	-	-	-	0.1	0.2	0.2	0.3	0.4	0.4
Performance rights	0.6	0.7	0.8	1.0	1.0	1.0	1.2	1.3	1.3	1.4
Synchronization revenues**	-	-	-	-	-	-	-	-	-	0.3
Total*	23.9	22.4	20.8	20.8	20.2	19.6	18.4	17.1	16.0	15.1

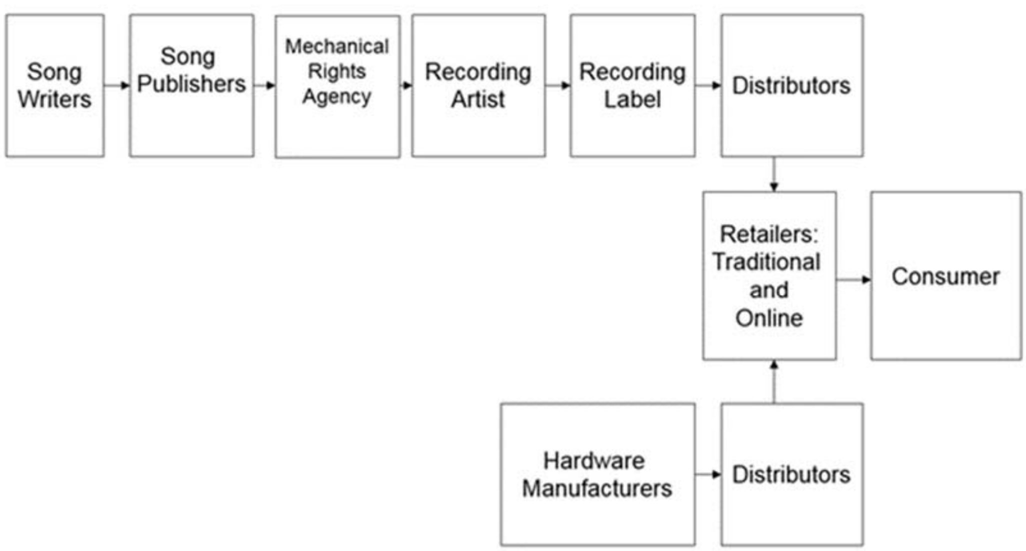
(In US$ billions)	2011	2012	2013	2014	2015	2016	2017	2018
Physical	8.3	7.7	6.8	6.0	5.8	5.6	5.2	4.7
Digital (excluding streaming)	4.3	4.4	4.3	4.0	3.8	3.2	2.6	2.3
Streaming	0.7	1.0	1.4	4.9	2.9	4.7	6.7	8.9
Performance rights	1.5	1.6	1.8	1.9	2.0	2.3	2.4	2.7
Synchronization revenues**	0.3	0.3	0.3	0.3	0.4	0.4	0.4	0.4
Total*	15.0	15.1	14.8	14.3	14.8	16.2	17.4	19.1

*Totals were different due to rounding

**Synchronization revenues: a music synchronization license, also known as a sync license, was a music license granted by the owner or composer of a particular piece of work.

Source: Created by the case authors using data from IFPI, *IFPI Global Music Report 2019–State of the Industry*, IFPI, April 2, 2019, 13, accessed May 6, 2019, www.ifpi.org/downloads/GMR2019.pdf.

EXHIBIT 2: MUSIC RECORDING INDUSTRY CHART



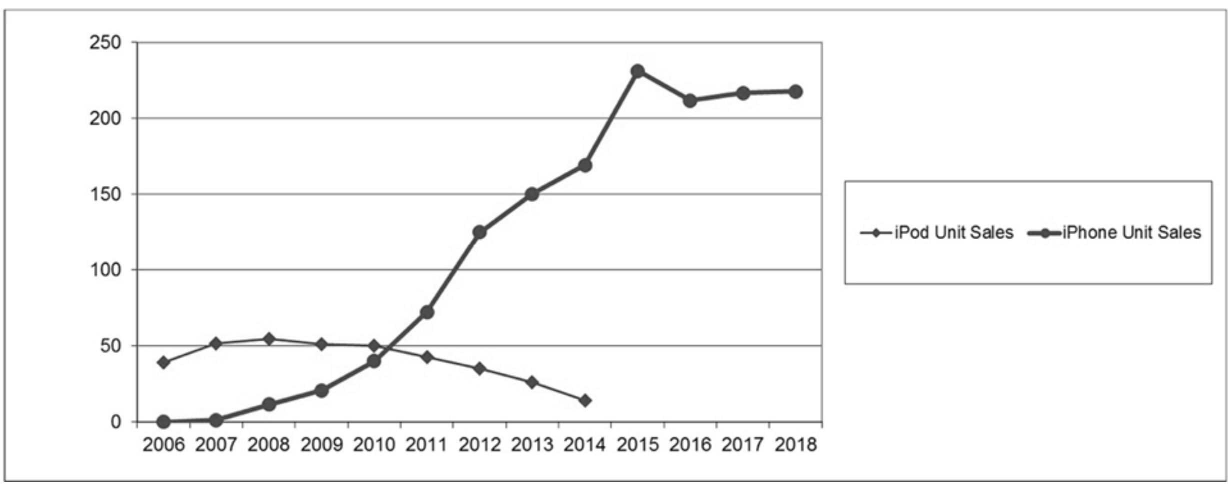
Source: Created by the case authors using data from various articles and industry executives.

EXHIBIT 3: VARIOUS ROYALTY AND REVENUE RATES FOR A TYPICAL CD RELEASE

Party	As % of Retail Price
Songwriters	0.3%
Song Publishers	0.3%
Mechanical license (for music reproduction)	3.5%
Recording artist-Gross Payments	16.5%
Less packaging costs	4.1%
Less promotional copies	1.2%
Less return reserves	2.9%
Less tours, videos, marketing	*Depends*
Less manager fees	3.3%
Recording artist-Net Payments	*Depends*
Distributors	10.0%
Retail	25.0%
Record label	*Depends*

Notes: Average CD Retail Price = $15.00; Recording artist net payments and record label gross profit varied depending on whether record sales could cover an estimated $600,000 in tours, video and marketing costs.
Source: Created by the case authors using data from various articles and industry executives.

EXHIBIT 4: IPOD AND IPHONE UNIT SALES, 2006–2018



Unit Sales (Millions)	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018
iPod Unit Sales	39.41	51.63	54.83	51.13	50.31	42.62	35.17	26.38	14.38				
iPhone Unit Sales	0	1.39	11.63	20.73	39.99	72.29	125.05	150.256	169.22	231.22	211.88	216.76	217.72

Notes: The iPhone was launched in 2007, and Apple stopped reporting iPod unit sales after 2014.
Source: Created by the case authors using data from Shanhong Liu, "Unit Sales of the Apple iPhone Worldwide from 2007 to 2018 (in Millions)," Statista, 2019, accessed May 6, 2019, www.statista.com/statistics/276306/global-apple-iphone-sales-since-fiscal-year-2007/; Statista Research Department, "Global Apple iPod Sales from 2006 to 2014 (in Million Units)," Statista, 2019, accessed May 6, 2019, www.statista.com/statistics/276307/global-apple-ipod-sales-since-fiscal-year-2006/.

EXHIBIT 5: APPLE MUSIC AND MEDIA REVENUE FORECAST

$ in Millions	2017	2018E	2019E	2020E	2021E	2022E	2023E	2024E	2025E
Installed Base of Apple Pay Devices	**1,420**	**1,531**	**1,621**	**1,738**	**1,856**	**1,963**	**2,076**	**2,192**	**2,321**
Y/Y % Change	9%	8%	6%	7%	7%	6%	6%	6%	6%
Apple Music Paid Subscriptions	**35**	**54**	**79**	**106**	**138**	**173**	**207**	**238**	**262**
Y/Y % Change	75%	55%	45%	35%	30%	25%	20%	15%	10%
% of Installed Base	2%	4%	5%	6%	7%	9%	10%	11%	11%
Apple Music Revenue - Domestic									
Number of Family Subscriptions	4	6	9	12	15	19	23	26	29
Average Subscription Price	$15	$15	$15	$15	$15	$15	$15	$15	$15
Family Subscription Revenue	$693	$1,073	$1,556	$2,101	$2,732	$3,415	$4,097	$4,712	$5,183
Number of Individual Subscriptions	14	21	31	41	54	67	81	93	102
Average Subscription Price	$10	$10	$10	$10	$10	$10	$10	$10	$10
Individual Subscription Revenue	$1,636	$2,536	$3,678	$4,965	$6,454	$8,068	$9,682	$11,134	$12,247
Total Domestic Revenue	$2,329	$3,610	$5,234	$7,066	$9,186	$11,483	$13,779	$15,846	$17,430
Apple Music Revenue - Abroad									
Number of Family Subscriptions	4	6	9	12	15	19	23	26	29
Average Subscription Price	$11	$11	$11	$11	$11	$11	$11	$11	$11
Family Subscription Revenue	$520	$806	$1,168	$1,577	$2,050	$2,563	$3,075	$3,537	$3,890
Number of Individual Subscriptions	14	21	31	41	54	67	81	93	102
Average Subscription Price	$7	$7	$7	$7	$7	$7	$7	$7	$7
Individual Subscription Revenue	$1,104	$1,710	$2,480	$3,348	$4,353	$5,441	$6,529	$7,508	$8,259
Total Abroad Revenue	$1,623	$2,516	$3,648	$4,925	$6,403	$8,004	$9,604	$11,045	$12,150
Total Apple Music Revenue	**$3,952**	**$6,126**	**$8,883**	**$11,992**	**$15,589**	**$19,486**	**$23,384**	**$26,891**	**$29,580**
Y/Y % Change	113%	55%	45%	35%	30%	25%	20%	15%	10%
Number of Media Subscriptions	-	-	-	5	10	17	26	36	46
Average Subscription Price (Incremental to Music)	$5	$5	$5	$5	$5	$5	$5	$5	$5
Media Subscription Revenue	$0	$0	$0	$319	$621	$1,035	$1,553	$2,143	$2,751
Apple Music + Media Revenue	**$3,952**	**$6,126**	**$8,883**	**$12,310**	**$16,210**	**$20,522**	**$24,937**	**$29,034**	**$32,331**
Y/Y % Change	113%	55%	45%	39%	32%	27%	22%	16%	11%

Source: Created by the case authors using data from Samik Chatterjee, “Apple,” J.P. Morgan, September 27, 2018, 100.

ENDNOTES

[1] This case has been written on the basis of published sources only. Consequently, the interpretation and perspectives presented in this case are not necessarily those of Apple Inc. or any of its employees.
[2] Tim Bradshaw, "Apple to Stop Revealing Unit Sales Sparking Peak iPhone Fears," *Financial Times*, November 1, 2018, accessed May 6, 2019, www.ft.com/content/5efb264c-de3d-11e8-9f04-38d397e6661c.
[3] Amy Watson, "Share of Music Streaming Subscribers Worldwide as of the First Half of 2018, By Company," 2019, accessed May 6, 2019, www.statista.com/statistics/653926/music-streaming-service-subscriber-share/.
[4] Rachel Rosmarin, "IPod Killers That Didn't",2006, accessed Feb 27, 2020, www.forbes.com/2006/10/20/ipod-zune-rio-tech-media-cx_rr_1023killers.html#490b246d1a91.
[5] Eliot Van Buskirk, "Zune Eats Creative's Meager Lunch, Grabbing 4 Percent of MP3 Player Market," 2008, accessed February 27, 2020, www.wired.com/2008/05/ipod-loses-mark.
[6] All dollar amounts in US dollars unless specified otherwise.
[7] "IFPI Global Music Report 2019: State of the Industry," IFPI, April 2, 2019, accessed May 6, 2019, www.ifpi.org/downloads/GMR2019.pdf.
[8] Ibid.
[9] "U.S. Recorded Music Revenues" Accesses Feb 27, 2020, www.riaa.com/u-s-sales-database/.
[10] Michael Mertens, *Thieves In Cyberspace: Examining Music Piracy And Copyright Law Deficiencies In Russia As It Enters The Digital Age*, 14, University of Miami International and Comparative Law Review, 139 (2006), accessed February 27, 2020, https://repository.law.miami.edu/umiclr/vol14/iss1/4.
[11] Mike Wiser, "Why Do CDs Cost So Much?," in "The Way the Music Died," PBS Frontline, May 27, 2004, accessed June 15, 2005, www.pbs.org/wgbh/pages/frontline/shows/music/inside/faqs.html.
[12] In 2011 UMG: 29.85%, SME 29.29%, WMG 19.13%, EMI 9.62%, in total these four have 87.89%. The Nielsen Company & Billboard's 2011 Music Industry Report. Businesswire, 2012, accessed February 27, 2020, www.businesswire.com/news/home/20120105005547/en/Nielsen-Company-Billboard%E2%80%99s-2011-Music-Industry-Report.
[13] Geoffrey Hull, Thomas Hutchison, and Richard Strasser, 2011. *The Music and Recording Business*. Routledge. (p 281).
[14] Sudip Bhattacharjee, Ram D. Gopal, Kaveepan Lertwachara and James R. Marsden, "Impact of Legal Threats on Online Music Sharing Activity: An Analysis of Music Industry Legal Actions," *The Journal of Law & Economics* Vol. 49, No. 1 (April 2006).
[15] "RIAA sues MP3.com, alleges copyright violations", CNET, January 2002, accessed February 28, 2020, /www.cnet.com/news/riaa-sues-mp3-com-alleges-copyright-violations/.
[16] Maria Seminerio,"Stop the music! Diamond Multimedia faces lawsuit over its Rio player," October 1998, accessed Feb 28, 2020, www.zdnet.com/article/stop-the-music-diamond-multimedia-faces-lawsuit-over-its-rio-player/.
[17] Sam Costello, "Digital Music Security Initiative Nearly Ready," CNN, September, 2000, accessed March 2, 2020, www.cnn.com/2000/TECH/computing/09/22/SDMI.prep.idg/index.html.
[18] Sue Zeidler,"Hackers Insist They Beat Audio Technology," CIOL, October 2000, accessed March 2, 2020, www.ciol.com/hackers-insist-beat-audio-technology/.
[19] Steven Levy, "The Man Can't Stop Our Music," *Newsweek*, March 27, 2000, accessed July 25, 2019, www.newsweek.com/man-cant-stop-our-music-156803.
[20] Edward Hellmore, "Music Industry Is Caught Napping," *Guardian*, March 16, 2000, accessed July 25, 2019, www.theguardian.com/technology/2000/mar/16/onlinesupplement4.
[21] "Napster Has a Plan," The Economist, February 2001, accessed February 28, 2020, www.economist.com/unknown/2001/02/21/napster-has-a-plan.
[22] Vickie L. Feeman, William S. Coats, Heather D. Rafter, and John G. Given, "Revenge of the Record Industry Association of America: The Rise and Fall of Napster," *Vill, Sports & Ent. LJ* 9 (2002): 35. (p 36, footnote 9).
[23] Sam Costello, "Anti-Napster Ruling Draws Mixed Reaction," PCWorld, February 12, 2001, accessed July 25, 2019, www.pcworld.idg.com.au/article/20087/anti-napster_ruling_draws_mixed_reaction/.
[24] Hellmore, op. cit. [Edward Hellmore, "Music Industry Is Caught Napping," *Guardian*, March 16, 2000, accessed July 25, 2019, www.theguardian.com/technology/2000/mar/16/onlinesupplement4.]
[25] Matt Richtel, "The Napster Decision: The Overview; Apellate Judges Back Limitations on Copying Music," *New York Time*, February 13, 2001, accessed February 29, 2020, https://www.nytimes.com/2001/02/13/business/napster-decision-overview-appellate-judges-back-limitations-copying-music.html; Peter Cohen "Roxio to Acquire Napster Assets," MawWorld, November 15, 2002, accessed February 29, 2020, https://www.macworld.com/article/1007839/roxio.html
[26] "Justice Dep't Begins Probe Into MusicNet, Pressplay," Billboard, August 2001, accessed February 29, 2020, www.billboard.com/articles/news/78850/justice-dept-begins-probe-into-musicnet-pressplay.
[27] "12-year-old Settles Music Swap Lawsuit," CNN, February 2004, accessed February 29, 2020, www.cnn.com/2003/TECH/internet/09/09/music.swap.settlement/.
[28] Wiser, op. cit. [Mike Wiser, "Why Do CDs Cost So Much?," in "The Way the Music Died," PBS Frontline, May 27, 2004, accessed June 15, 2005, www.pbs.org/wgbh/pages/frontline/shows/music/inside/faqs.html.]
[29] Amy Schatz et al., "High Court to Old Media: You Win," *Wall Street Journal*, June 28, 2005, B1.
[30] "About Piracy," RIAA, accessed July 26, 2019, www.riaa.com/resources-learning/about-piracy/.
[31] Tim Bradshaw, "Apple Wins Race to be First Trillion-Dollar Company," *Financial Times*, August 2, 2018, accessed May 6, 2019, www.ft.com/content/aebad290-9644-11e8-b67b-b8205561c3fe.

[32] Mike Schramm, "Dell Speaks Out on Infamous Apple Quote," engadget, October 18, 2011, accessed May 6, 2019, www.engadget.com/2011/10/18/dell-speaks-out-on-infamous-apple-quote/.
[33] Jeremy Reimer, "Personal Computer Market Share: 1975-2004," accessed February 29, 2020, www.retrocomputing.net/info/siti/total_share.html.
[34] Dennis Sellers, "Apple 'Manifesto': 5 Down, 95 To Go," May 2001, accessed February 29, 2020, www.macworld.com/article/1017497/manifesto.html.
[35] Each gigabyte could store approximately 1,024 megabytes. Each digital tune was, on average, four to five megabytes.
[36] "iPod," Computer Hope, accessed January 11, 2007, www.computerhope.com/jargon/i/ipod.htm.
[37] Steve Knopper," iTunes' 10th Anniversary: How Steve Jobs Turned the Industry Upside Down," Rolling Stone, April, 2013, accessed February 29, 2020, www.rollingstone.com/culture/culture-news/itunes-10th-anniversary-how-steve-jobs-turned-the-industry-upside-down-68985/.
[38] Ina Fried, "Will iTunes Make Apple Shine?," CNET News, October 16, 2003, accessed July 25, 2019, www.cnet.com/news/will-itunes-make-apple-shine/.
[39] Brian Osborne," iTunes reaches one billion song downloads" , Geek.com, February, 2006, accessed February 29, 2020, www.geek.com/apple/itunes-reaches-one-billion-song-downloads-561816/.
[40] "Tectonic Change In Music Industry", CBS News, May 2003, accessed February 29, 2020, www.cbsnews.com/news/tectonic-change-in-music-industry.
[41] ExtremeTech Staff, "DRM Stripped from iTunes Downloads," accessed July 25, 2019, www.extremetech.com/extreme/74798-drm-stripped-from-itunes-downloads.
[42] Tony Smith, "Apple Plugs PyMusique iTunes Hole," *The Register*, March 2005, accessed March 2, 2020, www.theregister.co.uk/2005/03/22/apple_blocks_pymusique/.
[43] "FairPlay," TechTarget, accessed March 24, 2020, https://whatis.techtarget.com/definition/FairPlay.
[44] Matt Graves, "Napster's Improbably Journey," Medium, June 15, 2016, accessed May 5, 2019, https://medium.com/cuepoint/napsters-improbable-journey-ac73e9915e6a.
[45] Ibid
[46] "Microsoft Unveils Zune Site-Changes MSN Music," CIO, November 2006, accessed February 29, 2020, www.cio.com/article/2443676/microsoft-unveils-zune-site--changes-msn-music.html.
[47] "Microsoft Zune: iPod Rival or Just Another MP3 Player?" CIO, November 2006, accessed February 29, 2020, www.cio.com/article/2443558/microsoft-zune--ipod-rival-or-just-another-mp3-player-.html.
[48] Laurie J. Flynn, "Music Player From Microsoft Offers Wireless Song-Sharing," *The New York Times*, September 2006, accessed February 29, 2020, www.nytimes.com/2006/09/15/technology/15soft.html.
[49] "Apple takes Podcasting Mainstream," Apple, June 28, 2005, accessed March 2, 2020, www.apple.com/newsroom/2005/06/28Apple-Takes-Podcasting-Mainstream/
[50] Brandon Griggs and Todd Leopold, "How iTunes Changed Music, and the World," CNN Business, April 26, 2013, accessed May 5, 2019, www.cnn.com/2013/04/26/tech/web/itunes-10th-anniversary/index.html.
[51] Shanhong Liu, "Unit Sales of the Apple iPhone Worldwide from 2007 to 2018 (in Millions)," Statista, 2019, accessed May 6, 2019, www.statista.com/statistics/276306/global-apple-iphone-sales-since-fiscal-year-2007/.
[52] Statista Research Department, "Global Apple iPod Sales from 2006 to 2014 (in Million Units)," Statista, 2019, accessed May 6, 2019, www.statista.com/statistics/276307/global-apple-ipod-sales-since-fiscal-year-2006/.
[53] Luke Dormehl, "Rest in Pod: Apple No Longer Reporting iPod Sales," *Cult of Mac* (blog), January 30, 2015, accessed May 6, 2019, www.cultofmac.com/310636/rest-pod-apple-no-longer-reporting-ipod-sales/.
[54] Richard Smirke, "IFPI Global Report 2018: Music Sales Rise for Third Straight Year as Streaming Becomes Biggest Revenue Source," Billboard, April 2018, accessed March 2, 2020, www.billboard.com/articles/business/8372798/ifpi-global-report-2018-music-sales-rise-streaming-revenue.
[55] Sarah Perez, "Streaming Accounted for Nearly Half of Music Revenues Worldwide in 2018," Techcrunch, April 2019, accessed February 29, 2020, https://techcrunch.com/2019/04/02/streaming-accounted-for-nearly-half-of-music-revenues-worldwide-in-2018/.
[56] Sarah Perez, "Streaming Services Accounted for Nearly 80% of All Music Revenue in 2019," accessed February 29, 2020, https://techcrunch.com/2020/02/26/streaming-services-accounted-for-nearly-80-of-all-music-revenue-in-2019/.
[57] Michael Grothaus, "Apple Music Now Has More Than 10 Million Paying Subscribers, *Fast Company*, January 16, 2016, accessed March 24, 2020, www.fastcompany.com/3055328/apple-music-now-has-more-than-10-million-paying-subscribers; Lucy England, "An Apple Exec Just Revealed How Much Apple Music is Really Going to Pay Record Labels," *Business Insider*, June 16, 2015, accessed March 24, 2020, www.businessinsider.com/revenue-split-for-apple-music-2015-6.
[58] "Apple Music Passes 50 Million Subscribers, Including Free Trials," *Variety*, May 15, 2018, accessed March 24, 2020, https://variety.com/2018/biz/news/apple-music-passes-50-million-subscribers-including-free-trials-1202811061/.
[59] Mitchel Broussard, "Spotify Now Has 83M Paid Subscribers and 180M Total Monthly Users," Mac Rumors, July 2018, accessed March 2, 2020, www.macrumors.com/2018/07/26/spotify-83m-paid-subscribers/.
[60] Mintel, Streaming Audio – US – October 2018, Mintel Group Ltd. 2–4.
[61] Adam Rogers, "US Music Streaming: Apple Music and Spotify Dominate," December 2019, accessed March 2, 2020, https://articles2.marketrealist.com/2019/12/us-music-streaming-apple-music-and-spotify-dominate/.
[62] David Paddon, "Apple's Subscription News Service Has Several Canadian Media Partners at Launch," *Financial Post*, March 25, 2019, accessed May 17, 2019, https://business.financialpost.com/pmn/business-pmn/apple-news-plus-subscription-service-has-several-canadian-media-partners-at-launch.